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Time-Dependent Density Functional Methods for Raman Spectra in Open-Shell Systems

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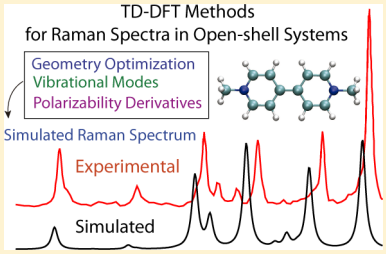
Supporting Information

ABSTRACT: We present an implementation of a time-dependent density functional theory (TD-DFT) linear response module in NWChem for unrestricted DFT calculations and apply it to the calculation of resonant Raman spectra in open-shell molecular systems using the short-time approximation. The new source code was validated and applied to simulate Raman spectra on several doublet organic radicals (e.g., benzyl, benzosemiquinone, TMPD, *trans*-stilbene anion and cation, and methyl viologen) and the metal complex copper phthalocyanine. We also introduce a divide-and-conquer approach for the evaluation of polarizabilities in relatively large systems (e.g., copper phthalocyanine). The implemented tool gives comparisons with experiment that are similar to what is commonly found for closed-shell systems, with good agreement for most features except for small frequency shifts, and occasionally large deviations for some modes that depend on the molecular system studied, experimental conditions not being accounted in the modeling such as solvation effects and extra solvent-based peaks, and approximations in the underlying theory. The approximations used in the quantum chemical modeling include (i) choice of exchange-correlation functional and basis set; (ii) harmonic approximation used in the frequency analysis to determine vibrational normal modes; and (iii) short-time approximation (omission of nuclear motion effects) used in calculating resonant Raman spectra.



INTRODUCTION

Raman spectroscopy is a tool to probe vibrational modes in a molecular system via inelastic scattering of monochromatic electromagnetic waves at visible, near-infrared, or near-ultraviolet wavelengths. Each molecular structure has a unique Raman spectrum that provides a signature that is useful for sensing applications. Quantum chemical modeling of a Raman spectrum offers deep insight to the intricate mechanism in which molecules interact with two photons, and the results are often useful for assigning and interpreting experimental spectra.[1,2] If the energy of a photon that excites a molecule is not resonant with some molecular electronic transition, the normal Raman (NR) spectrum is obtained. However, if the energy is resonant with some molecular (usually vibronically allowed) transition, the resonant Raman (RR) spectrum[3] is obtained. Since RR intensities are often 10^4–10^6 stronger than NR, there is much interest in RR measurements, yet it is common to use NR calculated spectra to interpret the results as methods for calculating RR spectra are not commonly available. Existing first principles quantum chemical simulations based on density functional theory (DFT) methodology have been presented many times for NR spectra, such as in the work of Chen et al.[4] who used real-time time-dependent density functional theory (RT-TDDFT) based on the CP2K code[5] for the closed-shell molecule pyridine. Another example of NR spectra is the work of Johnson et al.[6] as implemented in QChem,[7] which can handle NR spectra for closed/open-shell molecular systems but only in the static (zero frequency limit). The ability to describe both NR and RR spectra was developed by Jensen et al.[8−10] using time-dependent DFT (TD-DFT) with the AOResponse module in ADF.[11]This approach only considers closed shell molecular systems. ADF uses Slater-type orbitals (STOs) and has a limited set of exchange-correlation functionals. Finally, Mullin et al.[12,13] adapted the AOResponse module[14−16] for NWChem[17] but again only for restricted (i.e., closed-shell) DFT calculations. NWChem uses Gaussian-type orbitals (GTOs), and it provides a wide range of exchange-correlation functionals, so this work explored the effect of using several density functionals and basis sets in the simulation of the Raman spectrum of pyridine, including studies of the chemical contribution to the SERS enhancement factor.[18,19] In the present work we add new features to the NWChem Raman code, enhancing the AOResponse module to calculate normal/resonance Raman spectra for open shell systems. Also, we introduce a *Divide-and-Conquer* scheme in the evaluation of polarizabilities that lets us study relatively large molecular systems (e.g., copper phthalocyanine). This implementation also enables the study of many other properties, including NMR chemical shieldings[20,21] on nonrelativistic and relativistic (closed/open) shell systems, hyperfine coupling constants on nonrelativistic and relativistic open shell systems[22] and *g*-shifts on relativistic open shell systems[22] for a broad range of radicals and, in general, molecular systems that have inherent magnetic properties due to open-shell character.

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This article is organized as follows: in section 2 we give a detailed description of the work done to improve the NWChem Raman simulator (AOResponse module). Section 3 presents benchmarks for a set of radical doublets where simulated Raman spectra are compared side-by-side with the corresponding experimental Raman spectra. Finally, section 4 gives a summary, conclusions, and recommendations for future improvements related to this work.

■ COMPUTATIONAL AND IMPLEMENTATION DETAILS

The simulation of Raman spectra, within DFT, consists of several steps (see Figure 1): (i) geometry optimization, (ii)

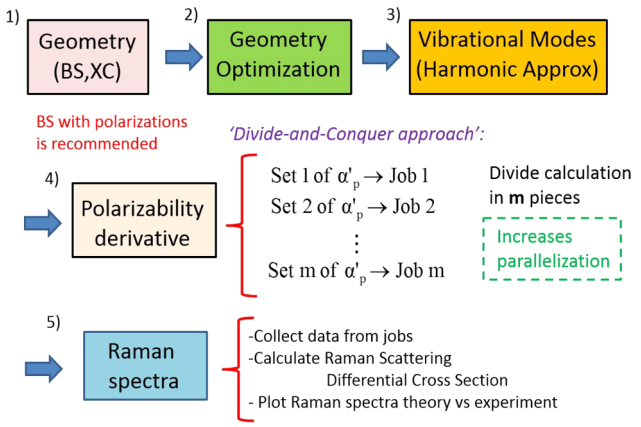

Figure 1. Flow diagram of Raman simulation: (1) Input geometry of molecular system and choose a proper basis set (BS) and exchange-correlation functional (XC). (2) Obtain an optimized geometry. (3) Calculation of normal vibrational modes using harmonic approximation. (4) Computation of polarizability derivatives using *Divide-and-Conquer* approach. (5) Calculation of Raman spectra.

calculation of vibrational normal modes,[23] (iii) calculation of polarizabilities, and (iv) obtaining Raman spectra. The present section will be concerned with steps (iii) and (iv) because these involve the new developments presented in this work.

Raman intensities are expressed in terms of differential scattering cross sections, $d\sigma_p/d\Omega$, which are calculated using

$$\frac{d\sigma_p}{d\Omega}\left(\theta = \frac{\pi}{2}, \omega\right) = \frac{\pi^2}{\varepsilon_0^{\,2}}(\omega - \omega_p)^4 \frac{h}{8\pi^2 c\omega_p}[45\alpha_p'^2 + 7\gamma_p'^2] \frac{1}{45(1 - \exp(-hc\omega_p/k_B T))} \left[\frac{m^2}{sr}\right] \quad (1)$$

where $\theta = \pi/2$ denotes the relative angle between the incident laser beam and location of detector, ω and ω_p are the frequencies of the incident and pth normal mode, respectively. The scattering factor, $45\alpha_p'^2 + 7\gamma_p'^2$, is expressed in terms of the isotropic polarizability derivative, α_p', and the anisotropic polarizability derivative, γ_p'. These are expressed as:

$$\alpha_p' = \frac{1}{3}\{(\bar{\alpha}_{xx}')_p + (\bar{\alpha}_{yy}')_p + (\bar{\alpha}_{zz}')_p\} \quad [C\,m\,kg^{-1/2}\,V^{-1}] \quad (2)$$

$$\gamma_p'^2 = \frac{1}{2}\{[(\bar{\alpha}_{xx}')_p - (\bar{\alpha}_{yy}')_p]^2 + [(\bar{\alpha}_{yy}')_p - (\bar{\alpha}_{zz}')_p]^2 + [(\bar{\alpha}_{zz}')_p - (\bar{\alpha}_{xx}')_p]^2 + 6[(\bar{\alpha}_{xy}')_p^{\,2} + (\bar{\alpha}_{yz}')_p^{\,2} + (\bar{\alpha}_{zx}')_p^{\,2}]\} \quad [C^2\,m^2\,kg^{-1}\,V^{-2}] \quad (3)$$

with the polarizability derivative given as

$$(\bar{\alpha}_{uv}')_p = \left(\frac{\partial\bar{\alpha}_{uv}}{\partial R_p^{(q)}}\right)_0 \quad [C\,m\,kg^{-1/2}\,V^{-1}] \quad (4)$$

Here u,v = x,y,z and $R_p^{(q)}$ are the mass-weighted normal coordinates. Equation 4 is evaluated numerically using a two-point formula in what we present. The simulated Raman spectra are obtained from eq 1, in which the stick spectrum is broadened by a Lorentzian function with a 20 cm^{-1} width. To calculate frequency-dependent polarizabilities. we used the so-called Short-Time Approximation[8,24−28] in which finite lifetime effects are included in the electronic response, but the nuclear positions are frozen.[29] An earlier version of the Raman module that simulates Raman spectra in NWChem was developed and benchmarked by Mullin et al.[12,13] This only does simulations on closed-shell systems since the implemented polarizability methodology was limited to computing perturbed electronic densities for that type of system. In the present work, we revised/improved the source code to compute perturbed electronic densities, extending its capability to deal with open shell systems too. The evaluation of perturbed electronic densities is done within TD-DFT and involves solving self-consistently the coupled-perturbed Kohn−Sham[29,30] (CPKS) equations whose convergence rate to a solution has been improved as shown in Table 1. In the convergence test we used

Table 1. Comparison of convergence rate between Old CPKS and New CPKS implementations

Old CPKS implementation		New CPKS[a] implementation	
Iter Nr	residual[b]	Iter Nr	residual[b]
1	2.85×10^{00}	1	2.85×10^{00}
2	1.72×10^{00}	2	1.72×10^{00}
3	3.13×10^{-01}	3	3.14×10^{-01}
4	1.26×10^{-01}	4	1.27×10^{-01}
	...		...
26	2.99×10^{-03}	21	4.15×10^{-06}
27	7.86×10^{-03}	22	2.40×10^{-06}
28	2.10×10^{-02}	23	7.40×10^{-07}

[a]New CPKS implementation was released on NWChem trunk on June 2012. [b]The CPKS equations can be portrayed as solving self-consistently a standard linear equation **Ax** = **b**. A measure of how well one approaches the solution is given by computing the residual **r**, which is defined as **r** = **Ax** − **b**.

the chiral transition metal complex $[Co(en)_3]^{3+}$ studied in ref 31. The improved convergence is based on the following improvements to the CPKS module (the AOResponse module): (i) modifying the source code toward object-oriented design; (ii) reducing memory costs; (iii) improving the computational scheme to speed up the calculations (see below); (iv) adding a feature to read/save CPKS data, which can be used in a second calculation if needed; and (v) using complex linear algebra (e.g., calculations with complex global arrays (GA)[32]) when adding damping in the CPKS equations. Optionally to reduce computational effort in calculating Raman

spectra for large systems (e.g., CuPc metal complex with 57 atoms) we devised a computational scheme that divides the polarizability calculations into segments of tasks as shown in Figure 1. This scheme can be used for open- and closed-shell systems. In the Supporting Information, a more detailed explanation of how to do this type of calculation (also called "Recipe 3") is provided. We implemented additional keywords in the NWChem Raman module for that purpose.

In the next section we provide benchmarks of the implemented module that simulates Raman spectra in open-shell systems. We employed the Becke88[33] + Perdew86[34] (BP86) exchange-correlation functional, which shows moderate accuracy in predicting Raman spectra according to previous studies (e.g., Mullin et al.[12] and Jensen et al.[8]). The basis set was valence triple-ζ with polarization specially designed for density functional calculations (TZVP-DFT orbital) taken from the EMSL Basis Set Exchange.[35,36] For the case of the copper atom in copper phthalocyanine (CuPc), we used a basis set called def2-TZVP,[37] which has better quality than standard TZVP. We sometimes used in our benchmarks a basis set called Z2Pol,[38−40] which is a basis set designed for molecular electrical properties, to study its performance. Most of the simulated Raman spectra correspond to RR spectra. In order to determine the laser excitation wavelength on resonance we simulated the absorption spectra of the molecular system using oscillator strengths computed via TD-DFT. We identified the resonant excitation wavelength from maxima in the smoothed absorption spectrum that were similar to the experimental wavelengths.

■ RESULTS AND DISCUSSION

In this section, we present a validation of our implementation of linear-response theory for simulating Raman spectra of open-shell systems in NWChem by comparing simulation results against corresponding experimental spectra for a number of molecules. Note that we have verified that the code gives the same results for closed-shell molecules as the closed-shell version of AOResponse. Since the RR spectra of larger open-shell molecules have not been presented at the same level of theory, we have needed to use comparisons with experiment to provide a calibration of the results. Unfortunately this comparison is subject to a number of errors and uncertainties (sometimes in the experiments) as we shall see. In a few cases, we are able to compare our results with previous theory estimates of RR spectra obtained from simpler electronic structure models.

The following molecular systems were studied (Figure 2, obtained using VMD[41]): (1) Benzyl radical (C_7H_7). This is a simple example of an aromatic free radical that is very important as a reaction intermediate and is well studied experimentally and theoretically. (2) Benzosemiquinone radical anion (OC_6H_4OH-). This is a redox intermediate. (3) *N,N,N′,N′*-tetramethyl-*p*-phenylenediamine (TMPD) also called Wurster's blue.[42] This is an easily oxidized phenylenediamine that loses two electrons in one-electron oxidation steps. (4) *trans*-Stilbene (TS) anion and cation. *trans*-Stilbene is a hydrocarbon that consists of a *trans*-ethene double bond substituted with a phenyl group on both carbon atoms of the double bond; it is more stable than its corresponding isomer *cis*-stilbene. The TS geometry was initially taken from ref 43 and later optimized. (5) Methyl viologen (MV). This reaction product of methyl viologen dichloride is also called paraquat and is widely used as a herbicide. (6) Copper phthalocyanine

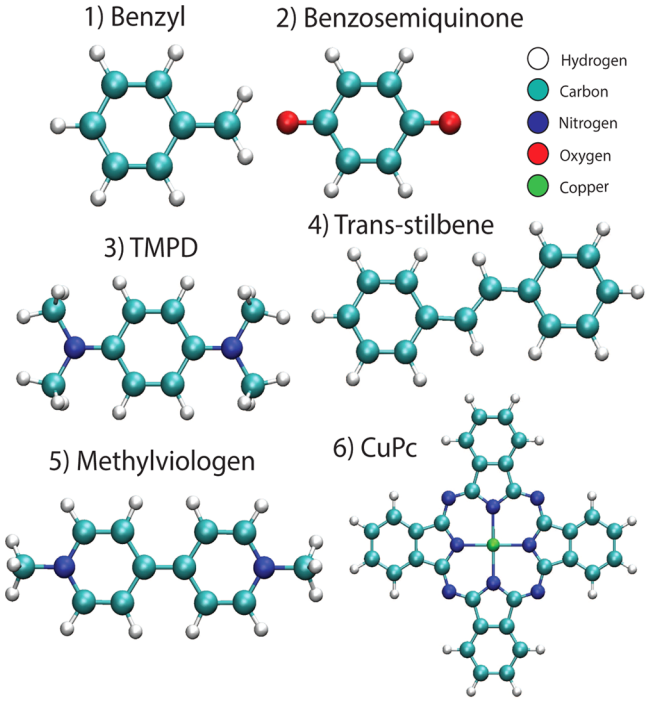

Figure 2. Geometries of molecular systems that were studied: (1) benzyl radical (C_7H_7); (2) benzosemiquinone radical anion (OC_6H_4O-); (3) *N,N,N′,N′*-tetramethyl- *p*-phenylenediamine (TMPD); (4) *trans*-stilbene anion and cation (TS); (5) methyl viologen (MV) cation; and (6) copper phthalocyanine (CuPc). All the systems are doublets.

(CuPc). CuPc is one type of p-type semiconductor that has interesting optical and electrical properties, has high thermal and chemical stability, and can be used as a thin film in advanced technological devices. It is also being used in inks among other applications. All the molecules studied are doublets.

For our simulations, the finite lifetime of the electronic excited states is introduced using a phenomenological damping parameter Γ. Γ values are in general system dependent.[44,45] However, as a good approximation that works for a broad range of molecules and which is compatible with the short time approximation, we have used $\Gamma = 0.004$ au as suggested by Jensen et al.[8,29]

The performance of our approach depends on several factors: (i) energy level structure of the molecule being studied; (ii) the exchange-correlation functional and basis set chosen; (iii) accuracy of the harmonic approximation used in predicting vibrational normal modes; (iv) accuracy of the short-time approximation used in computing polarizability derivatives that is related to the magninude of the Raman intensity peaks.

All the Raman spectra simulations are done in the gas phase, while the experimental Raman spectra have environmental influence (both static and dynamics) due to solvents. Also there are usually restrictions on the measurements such as the excitation wavelengths that are available. In the Supporting Information we show an example of a solvent effect on Raman spectra, which produces shifting of the Raman spectra. Our simulations, indeed, when compared against experiment indicate that shifting is important.

The threshold used in convergence of the CPKS equations for obtaining the perturbed density matrix was (1) 1×10^{-5} in benzyl; (2) 1×10^{-6} in benzosemiquinone; (3) 1×10^{-5} in

TMPD; (4) 1×10^{-5} in *trans*-stilbene anion and cation, respectively; (5) 5×10^{-5} in methyl viologen; and (6) 1×10^{-3} in copper phthalocyanine. In the case of CuPc we raised the threshold due to computational limitations with the relatively large system. However, we believe this limitation does not affect the quality of our results.

Test 1: Benzyl. Figure 3 shows the gas phase RR spectra of the benzyl radical using the BP86 exchange-correlation functional and TZVP basis set. The excitation wavelength for this choice of basis set and exchange-correlation functional is 360.7 nm. We compare this simulation of the Raman spectrum against an experimental time-resolved RR spectrum in methanol solution at 315 nm taken from Langkilde et al.[46] The features shown between 700−1000 cm^{-1} and 1400−1700 cm^{-1} of the simulated and experimental spectra are very similar except for a scaling factor. However, the experimental ratio is $I_{1150}/I_{1270} \approx 0.2$, while the simulation shows $I_{1150}/I_{1270} \approx 1.8$. There are a number of factors that can lead to this behavior, but such a drastic change in ratio of intensity peaks can hardly be due to solvent effects,which mostly produce minor change on intensity peaks. Other possible factors include the possibility that the experimental Raman spectrum is not really on-resonance. To study this, in the Supporting Information we show Raman simulations at two other wavelengths: 266 and 384.2 nm (Figure S2). We compare these results against the experimental Raman spectrum obtained using 315 nm laser excitation. No systematic improvement is demonstrated. Finally, an important possibility is that these modes are coupled by Fermi resonance. Indeed a simple visual examination of the normal mode eigenvectors shows that these two modes are superpositions of ring CH bend motions and the C−C stretch that connects the phenyl ring with the methylene side-chain. Since the C−C stretch is resonance stabilized for the benzyl radical, it is easy to imagine that a slight amount of anharmonicity would dramatically change the relative intensities of these two modes.

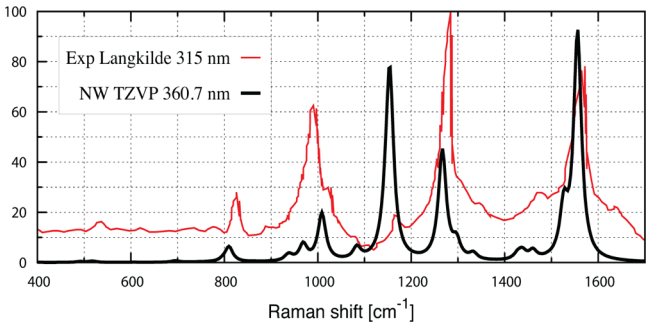

Figure 3. Gas phase simulation of RR spectrum of benzyl radical (doublet) excited at 360.7 nm using BP86 exchange-correlation functional and TZVP basis set. Experimental time-resolved RR spectrum of benzyl radical in methanol solution at 315 nm excitation wavelength is provided for comparison.[46]

Test 2: Benzosemiquinone. Figure 4 shows gas phase simulation of the Raman spectrum of doublet benzosemiquinone anion radical ($C_6H_4O_2$−) using the BP86 exchange-correlation functional and TZVP basis set at 417 nm excitation wavelength. In the same figure we show an experimental RR spectrum of the benzosemiquinone anion radical excited at 413.1 nm taken from Hester et al.[47] We have also replotted a shifted version of our simulated spectrum on top of the experimental spectrum to make comparisons. The results show (i) a 70 cm^{-1} downshift in the calculated spectrum leads to good alignment of most modes with experiment; presumably this shift is due to solvent effects; (ii) the relative intensities of most calculated peaks correspond reasonably with experiment; those that do not match are subject to Fermi-resonance effects similar to our discussion above for the benzyl radical but more complex because there are more coupled modes. Our simulated spectrum and the Hester et al. spectrum disagree with Tripathi's experimental results for the benzosemiquinone radical anion[48] as shown in Figure S3 (Supporting Information). The Tripathi results correspond better to the Raman spectrum of the benzosemiquinone neutral ($C_6H_4O_2H$) instead of benzosemiquinone radical anion ($C_6H_4O_2$−) as also shown in Figure S3.

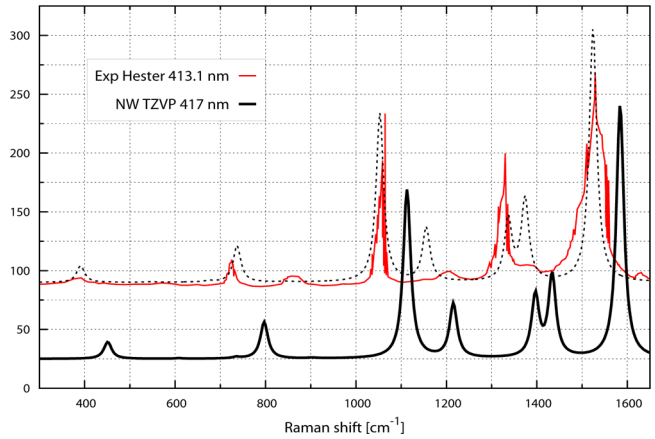

Figure 4. Gas phase simulations of off-resonance Raman spectrum of a doublet benzosemiquinone radical anion ($C_6H_4O_2$−) using BP86 exchange-correlation functional and TZVP basis set excited at 417 nm. Experimental RR spectrum of benzosemiquinone anion radical in ethanol solution excited at 413.1 nm excitation wavelength is provided for comparison.[47]

Test 3: TMPD. Figure 5 shows the gas phase simulation of the resonance Raman spectrum of the *N,N,N′,N′*-tetramethyl-*p*-phenylenediamine (TMPD) radical cation using the BP86

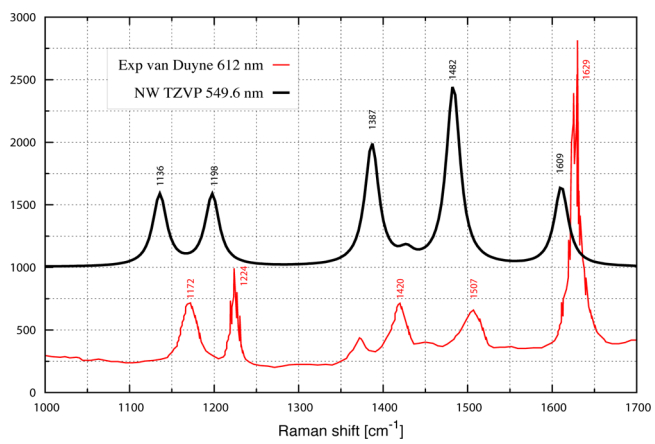

Figure 5. Gas phase simulation of RR spectrum of *N,N,N′,N′*-tetramethyl-*p*-phenylenediamine (TMPD) radical cation (doublet) excited at 549.6 nm using BP86 exchange-correlation functional and TZVP basis set. Experimental time-resolved RR spectrum of TMPD radical cation in 0.10 M TBAP/CH_3 (TBAP, tetrabutylammonium perchlorate) excited at 612 nm excitation wavelength is provided for comparison.[3]

exchange-correlation functional and TZVP basis set. The calculated resonance wavelength is 549.6 nm, which is closest to the wavelength 612 nm used in a time-resolved resonance Raman study of TMPD in 0.10 M TBAP/CH_3 (TBAP, tetrabutylammonium perchlorate).[3] The simulated Raman spectrum shows close similarity to the experimental spectrum except for a relative shifting caused by solvent effects. The structure of the simulated Raman spectrum between 1300−1700 cm^{-1} can be easily related to the experimental spectrum between 1400−1700 cm^{-1}, but the relative ratio of Raman peaks is not the same. Also there is a peak at 1375 cm^{-1} that does not appear in the simulation (see further discussion below). Similarly, the simulated Raman spectrum between 1100−1250 cm^{-1} can be related to the experimental Raman spectrum between 1150−1300 cm^{-1}. In the Supporting Information we present another experimental resonance Raman spectrum obtained with a laser excitation wavelength of 647.1 nm in methanol[49] (Figure S3). In this spectrum we can observe that the Raman intensity peak at 1375 cm^{-1} disappears, so presumably this is a solvent peak in the earlier result. We also show the simulated Raman spectrum using the Z2Pol basis set at a resonant wavelength (for that basis) of 555.8 nm. This spectrum has similar features to that obtained using TZVP but the shifting of the whole spectrum relative to experiment is less pronounced.

Test 4: *trans*-Stilbene Anion and Cation. Figure 6 shows the gas phase simulation of the Raman spectrum of the *trans*-stilbene (TS) radical cation using the BP86 exchange-correlation functional and TZVP basis set. We show two results: (i) the RR spectrum excited at 410.9 nm and (ii) the NR spectrum excited at 480 nm, which is rescaled to be compared to simulated RR spectrum. For comparison we present the experimental time-resolved RR spectrum of the TS radical cation using an excitation wavelength of 480 nm in 9,10-dicyanoanthracene solvent.[50] The simulated and experimental resonance Raman spectra compare very well in most features. However, the relative ratio of the two main intensity peaks does not match; experimentally $I_{1560}/I_{1290} \approx 2$, while in the simulated resonance Raman spectrum we get $I_{1590}/I_{1275} \approx 0.6$. This comparison is only slightly improved in the simulated normal Raman spectrum where we get $I_{1590}/I_{1275} \approx 0.9$. In the simulated resonance Raman spectrum we observe weak additional bumps from vibrational modes at 1090, 1175, and 1425 cm^{-1} that are not seen in the experiment. In the same figure we show a quantum chemical simulation of resonance Raman spectrum of *trans*-stilbene cation (TSC) from Negri et al.[51] who calculates equilibrium structures and vibrational force fields of the ground state using density functional theory (DFT). They employed the Becke exchange functional[33] and Lee, Yang, and Parr correlation functional[52] (BLYP) together with the Pople basis set 6-31G*. To calculate the resonant state they did an excited state calculation with a modified version of Quantum Consistent Force Field for π electrons (QCFF/PI).[53−55] The Negri et al. simulation shows the same features as in our simulation. Between 1150 and 1650 cm^{-1} we observe a shifting of the Raman intensity peaks relative to ours that could be due to the usage of the BLYP (instead of BP86) exchange-correlation functional in their work. In the Supporting Information we provide complementary experimental information by showing time-resolved resonance Raman spectrum at 485.3 nm excitation wavelength in 9,10-dicyanoanthracene solvent[56] (Figure S4).

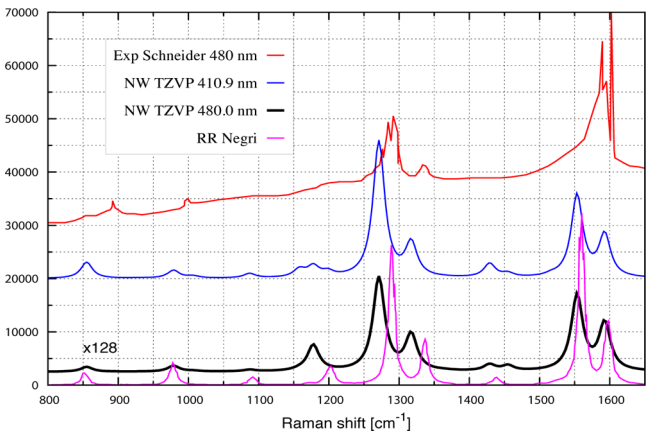

Figure 6. Gas phase simulations of Raman spectrum of *trans*-stilbene (TS) radical cation (doublet) using BP86 exchange-correlation functional and TZVP basis set: (i) RR spectrum excited at 410.9 nm and (ii) NR spectrum excited at 480 nm. Experimental time-resolved RR spectrum of TS radical cation is provided for comparison excited at 480 nm excitation wavelength in 9,10-dicyanoanthracene solvent.[50] Also in the graph is resonance Raman simulation from Negri et al.[53]

Figure 7 shows our simulation of the Raman spectrum of the *trans*-stilbene (TS) radical anion using the BP86 exchange-correlation functional and TZVP basis set. Similarly to the previous case of TS radical cation we present two simulations: (i) RR spectrum at 431.5 nm and (ii) NR spectrum at 480 nm. The experimental time-resolved RR spectrum of the TS radical anion excited at 480 nm in amine solvent is shown for comparison. The simulated resonance Raman spectrum does not agree very well with the corresponding experimental resonance Raman spectrum as there are extra Raman peaks in the calculated spectrum. However, the normal Raman spectrum excited at 480 nm shows better overall agreement with the experimental results, with the exception of showing a wrong

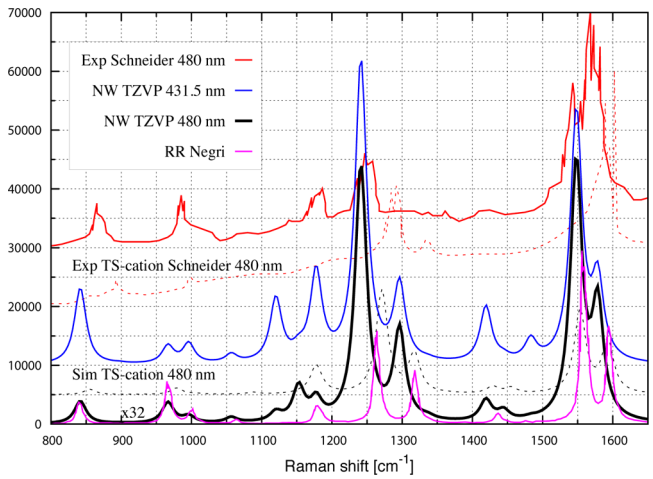

Figure 7. Gas phase simulations of Raman spectrum of *trans*-stilbene (TS) radical anion (doublet) using BP86 exchange-correlation functional and TZVP basis set: (i) RR spectrum excited at 431.5 nm and (ii) NR spectrum excited at 480 nm. Experimental time-resolved RR spectrum of TS radical anion is provided for comparison excited at 480 nm excitation wavelength in amine solvent. Dotted lines show experimental and simulated Raman spectra of TS radical cation at 480 nm excitation wavelength.[50] Also in the graph is a resonance Raman simulation from Negri et al.[53]

relative ratio of the two main Raman intensity peaks: in the simulation we have $I_{1550}/I_{1250} \approx 1$, while the experiment shows $I_{1575}/I_{1250} \approx 2.5$. In dashed curves in this graph, we added experimental and simulated Raman spectra for the cation TS to show that both behave similarly upon changing the charge of the molecular system (shifting to higher energy when charge is added). This comparison is also consistent with the older work of Schneider et al.[50] Finally, in the same graph we show results from Negri et al.[51] where the structure of their Raman spectrum is similar to ours, but their relative ratio of Raman intensity peaks is in better agreement with experiment than ours. In the Supporting Information we show the performance of the Z2Pol basis set in the simulation of TS anion Raman spectrum (Figure S5).

Test 5: Methyl Viologen. Figure 8 shows the gas phase normal Raman spectrum of the methyl viologen (MV) radical

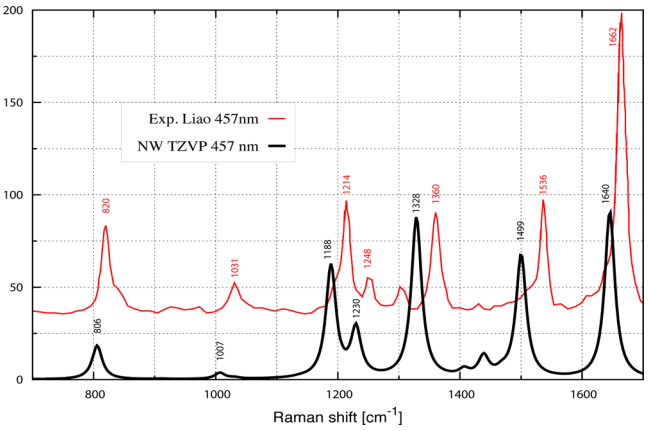

Figure 8. Gas phase simulation of the Raman spectrum of the methyl viologen (MV) radical cation (doublet) using the BP86 exchange-correlation functional and TZVP basis set excited at 457 nm. The experimental time-resolved RR spectrum of the MV radical cation (inside mesoporous MCM-410 at 457 nm is provided for comparison).[57]

cation using the BP86 exchange-correlation functional and TZVP basis set excited at 457 nm. We provide, for comparison, an experimental time-resolved RR spectrum of MV radical cation excited at 457 nm excitation wavelength inside mesoporous MCM-41.[57] We observe that the important normal vibrational modes appear in both simulation and experiment, and there is generally good agreement in the appearance of the spectra. However, the relative ratio of intensity peaks shows some differences. For example, the relative ratio of the two highest energy Raman intensity peaks $I_{1660}/I_{1540} \approx 2$ in the experiment and $I_{1640}/I_{1500} \approx 1.34$ in the calculations. In Figure 9 we show a RR simulation with the same functional and basis set at 535.5 excitation wavelength. These results compare very well with the experimental time-resolved RR spectrum at 530.9 nm excitation wavelength in water solution.[58] Additionally, we show in the same figure that using the Z2Pol basis set for simulating resonance Raman spectrum at 543.2 nm excitation wavelength produces results that are similar to the TZVP results. We also show a simulation of the Raman spectrum using the approach of Valley and co-workers[59] who use an indirect method to determine Raman spectra in open shell molecules. From the graph we can see that most of the features of our Raman simulation are reproduced with the exception of a wrong ratio of the two last intensity

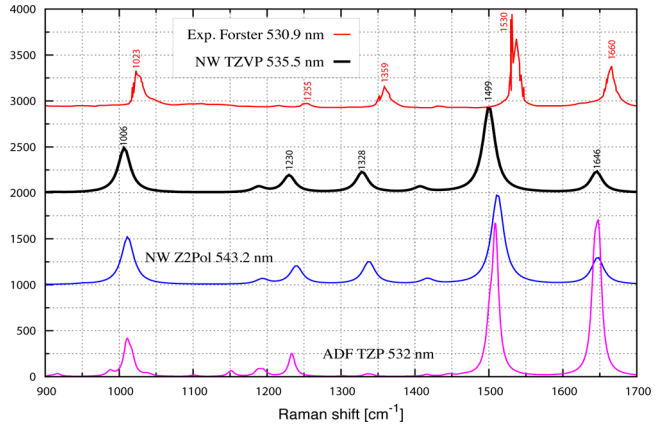

Figure 9. Gas phase simulations of Raman spectrum of methyl viologen (MV) radical cation (doublet) using BP86 exchange-correlation functional: (i) with TZVP basis set excited at 535.5 nm in NWChem; (ii) with Z2Pol basis set excited at 543.2 nm in NWChem, and (iii) with TZP basis set excited at 532 nm in ADF. Experimental time-resolved RR spectrum of MV radical cation is provided for comparison. It is excited at 530.9 nm excitation wavelength in water solution.[58]

peaks: the (NWChem,BP86,TZVP, 535.5 nm) result is $I_{1645}/I_{1500} \approx 0.25$, while the (ADF,BP86,TZP, 532 nm) gives $I_{1645}/I_{1500} \approx 1$. The corresponding experimental result is $I_{1660}/I_{1525} \approx 0.5$.

Test 6: Copper Pthalocyanine. Figure 10 shows two gas phase simulations of the Raman spectrum of copper

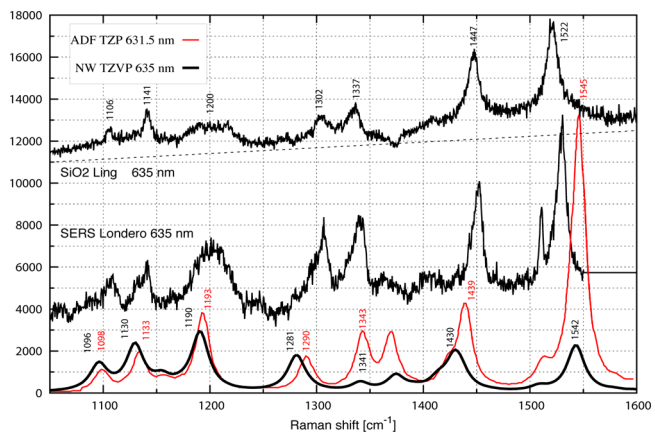

Figure 10. Gas phase simulations of the Raman spectrum of copper phthalocyanine (CuPc) (doublet) using the BP86 exchange-correlation functional: (i) with TZVP basis set and excited at 635 nm in the Gaussian MO based NWChem (this work) and (ii) with TZP basis set and excited at 631.5 nm using the Slater MO based ADF,[62] but using the indirect method of Valley.[59] Experimental RR spectra of CuPc are provided for comparison: (a) SERS/Ag excited at 635 nm excitation wavelength[61] and (b) excited at 635 nm in SiO_2/Si substrate.[60]

phthalocyanine (CuPc) using the BP86 exchange-correlation functional: (i) using NWChem with TZVP basis set for hydrogen, carbon, and nitrogen atoms and def2-TZVP for the copper atom, excited at 635 nm (our results) and (ii) using ADF with TZP basis set excited at 631.5 nm (where the indirect method mentioned above was used[59]).[57] Both curves are scaled to about the same intensity to show how the relative ratio of Raman intensity peaks differs. Comparing both

simulations we notice important differences in the appearance of the spectra. In the same figure we also show two experimental Raman spectra that closely correlate with each other. It is more convenient to compare our simulations against the experimental Raman spectrum excited at 635 nm in SiO_2/Si substrate[60] rather than the experimental Raman spectrum from surface enhanced raman spectroscopy (SERS)/Ag[61] because in our simulation we do not account for the silver surface that modifies the relative ratio of Raman intensity peaks while producing an overall enhancement of the spectra. We are assuming that the effect of the SiO_2/Si substrate on the spectrum is minor. On the graph is added a baseline below the CuPc Raman spectrum with SiO_2/Si substrate to indicate that the background increases as one moves to the right in the plot. This background needs to be removed in comparing theory and experiment. Comparison of the simulated spectra against the CuPc Raman spectrum in SiO_2/Si substrate shows (i) in the range 1050 to 1325 cm^{-1} there is a good correlation between experiment and simulation; (ii) the excited vibrational mode at 1337 cm^{-1} in the experimental curve appears attenuated at 1341 cm^{-1} in our simulation results; (iii) the excited vibrational modes at 1447 and 1522 cm^{-1} in the experimental curve appears well represented at 1430 and 1542 cm^{-1} in our simulation results, respectively. We get a relative ratio of Raman intensity peaks $I_{1430}/I_{1542} \approx 0.9$ compared with the experimental relative ratio $I_{1447}/I_{1522} \approx 0.7$. The agreement will be improved if we account for the fact that background noise raises the Raman intensity peak at 1552 cm^{-1} by $\sim 0.4 \times 10^4$ with respect to the other Raman intensity peak at 1447 cm^{-1} in the experimental Raman spectrum giving $I_{1447}/I_{1522}|_{\text{no bg noise}} \approx 0.8$. In Table 2 we show a summary of the vibrational modes with prominent Raman intensities in the experiment and two DFT simulations. In the Supporting Information we supplement these results by adding two more experimental spectra: (i) Tip Enhanced Raman Spectrum (TERS) on a silver surface excited at 632.8 nm excitation wavelength[62] and (ii) resonance Raman spectrum of a α-CuPc powder on a silver disk excited at 668 nm excitation wavelength[63] (Figure S6).

Table 2. Vibrational Modes Showing the Prominent Intensities in the CuPc Raman Spectrum on a SiO_2/Si Substrate and from Two Gas Phase DFT Simulations

exptl[a] (cm^{-1})	NW-DFT[b] (BP86,TZVP) (cm^{-1})	ADF-DFT[c] (BP86,TZP) (cm^{-1})
1106	1096	1098
1141	1130	1133
1200	1190	1193
1302	1281	1290
1337	1341	1343
1447	1430	1439
1522	1542	1545

[a]Ref 60. [b]This work. [c]Ref 62.

■ SUMMARY AND CONCLUSIONS

We have shown results from a new implementation of the NWChem AOResponse Raman simulator for studying open-shell systems and studied its performance by comparison to several experimental Raman spectra. Our results show that while the predicted Raman spectra are in good qualitative agreement with experiments, there are numerous quantitative differences, differences that are also typically found in studies of this sort for closed shell systems. The quality of the agreement varies depending on the molecular system studied, experimental conditions (e.g., presence of solvents) not taken into account in the simulation, and many factors of the simulation technology, including (i) type of exchange-correlation functional and basis set used in each step of the simulation (e.g., geometry optimization, frequency analysis, and calculation of polarizabilities); (ii) harmonic approximation used in frequency analysis for predicting vibrational normal modes; (iii) short-time approximation (omission of nuclear motion effects) used in evaluation of the polarizability and polarizability derivative. Our Raman gas phase simulations considered several doublet molecules using the BP86 exchange-correlation functional and TZVP basis. For benzyl, benzosemiquinone, and TMPD, we reproduced the main features of the experimental Raman spectra but the correct relative ratio of Raman intensities is not achieved. With *trans*-stilbene anion and cation the overall features of Raman spectra were accurately reproduced; however, the experimental resonance Raman spectra compare best with the simulated normal Raman spectra. Our NR results are consistent with a previous simulation from Negri et al.[51] For methyl viologen, the two gas phase Raman simulations at 457 and 535.5 nm showed good correlation with experimental Raman spectra though relative Raman intensities are not reproduced in the same way. Finally, for copper phthalocyanine we found good agreement with the experimental Raman spectrum on a SiO_2 substrate. Comparisons with SERS/Ag[61] (or TERS/Ag[62] in Supporting Information) were less precise, as makes sense given that we have modeled the gas-phase spectrum. Even for these surface spectra, our simulation results show relative good agreement with experimental results except for the vibrational mode at 1337 cm^{-1}, which is attenuated at 1341 cm^{-1} in our simulation. This work can be further enhanced by adding the following desired features: (i) solvent effects added in the calculations; (ii) speeding up convergence in current MO−CPKS NWChem module by finding a better algorithm than the Krylov Subspace Accelerated Inexact Newton (KAIN)[64] or by implementing an AO−CPKS module following the recipes from Ochsenfeld et al.[65−68] who has computed NMR chemical shieldings for thousands of atoms; (iii) sometimes including anharmonic terms in the frequency analysis part of the calculations of vibrational normal modes (e.g., using the VSCF NWChem module[17,69−71]); and (iv) revise the short-time approximation approach used in calculating the polarizabilities to include nuclear vibrational effects.[44]

■ ASSOCIATED CONTENT

Ⓢ Supporting Information

Additional information/data is provided regarding (i) solvent effect on Raman spectra; (ii) experimental and simulated Raman spectra of benzyl, TMPD, *trans*-stilbene, and copper phthalocyanine; (iii) NWChem Raman module. update of keywords; (iv) Raman spectra simulation recipes in NWChem; (v) optimized geometries of studied molecular systems. This material is available free of charge via the Internet at http://pubs.acs.org.

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Notes

The authors declare no competing financial interest.

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